Project Plan & Task List

The Code Theorists

Team GitHub Repository Link:

https://github.com/jason-widjasena/project-ai-demo

Initial Project Plan and Task List:

https://canva.link/99b1f5igtc0vgxd

Initial Project Plan and Task List:

https://docs.google.com/spreadsheets/d/12cC2bCGbygLqg6BM8GAdZDNgiaaRm0kaKFSGSo4OAIU/edit?usp=sharing

*Other information can be found on README.md

Our Members





王裕充 - 1820262257

Winrich

(CEO)

Strengths:

1. **Detail-Oriented**: Pays close attention to details ensuring accuracy, consistency, and quality.
2. **Disciplined**: Responsible and makes sure tasks meet designated deadlines.
3. **Basic Technical Experience**: Has some familiarity with programming and is willing to develop practical skills further.

Responsibilities:

1. Manage team deadlines and keep track of each member’s work.
2. Divides responsibilities and makes sure every member does their assigned tasks.
3. Review and check each member’s work to make sure they meet task requirements.



庄翔俊- 1820262246

Jason

(CIO)

Strengths:

1. **Methodical**: Does things carefully and systematically.
2. **Adaptability**: Capable of adapting to various unexpected turns of events.
3. **Disciplined**: Able to be punctual especially regarding time management.
4. **Coding Experience**: Has multiple coding experience for various programming languages.

Responsibilities:

1. Manage the team's GitHub repository, including organizing files and markdown content.
2. Keep track of each individual task progress to help ensure tasks are done on time.
3. Handle task submissions and make sure tasks are submitted on time.



张天龙- 1820262227
Michael
(Content Lead)

Strengths:

1. **Concept Simplification**: Able to simplify abstract concepts into practical explanations, examples, and exercises.
2. **Narrative Flow**: Able to structure well-organized outlines to ensure smooth and logical explanations.
3. **Content Clarity**: Able to break down intricate concepts into clear, digestible steps.

Responsibilities:

1. Plan the outline of explanations, examples, and exercises in each deliverable.
2. Review content and material to make sure they are clear and easy to understand.



彭世涛- 1820262252
Owen
(Visual Design Lead)

Strengths:

1. **Analytical skills**: Can break down complex problems into more manageable components.
2. **Open-mindness**: Can recognize and set aside personal prejudices.
3. **Self–Reflection**: Can step back, admit when an initial thought might be wrong, and adjust plans when presented with better facts.
4. **Logical Deduction**: Can find the best possible answer by weighing pros and cons when facing difficulties.

Responsibilities:

1. Plan the design and theme of the study guide, presentation slides, and video
2. Make sure that formatting and materials of each deliverable is easy to read and comprehend



洪建民- 1820262212

Raphael

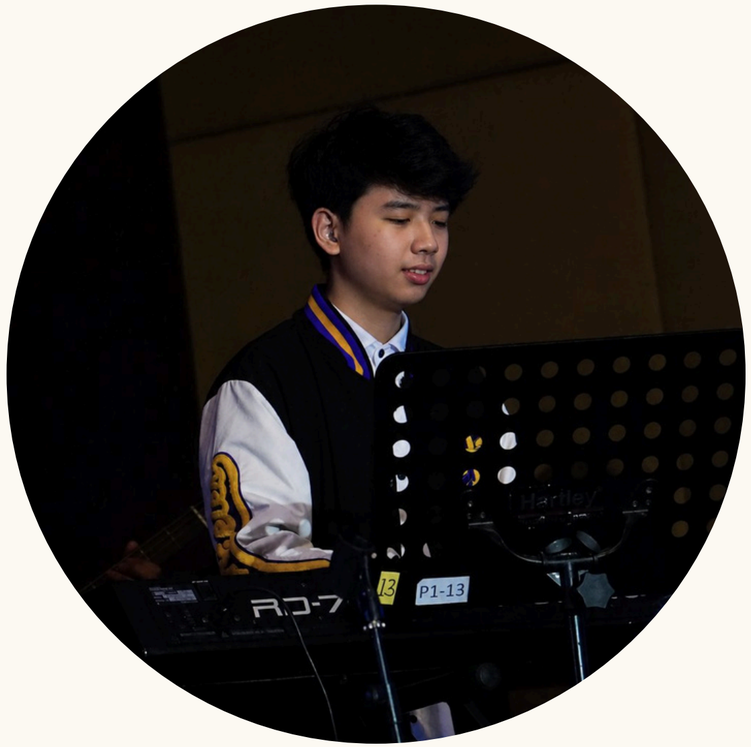
(Video Lead)

Strengths:

1. **Video Editing**: Experienced in using CapCut for video production.
2. **Problem Solving**: Approaches challenges analytically, breaking them down and working toward practical solutions.
3. **Eager to Learn**: Proactively explores new tools and skills, and applies them quickly.
4. **Adaptability**: Can adjust quickly to new environments, people, and expectations.

Responsibilities:

1. Plan and coordinate the production of the video, including theme planning, production, and editing.
2. Make sure all video content is engaging and well-organized.



潘超凡- 1820262213
Kenneth
(Quality Lead)

Strengths:

1. **Technical Experience**: Has experience in studying algorithms and data structures and in using Git workflows. Also has some experience programming languages, primarily Java, Kotlin, Go and TypeScript.
2. **Teaching Experience**: Has the skills to break down intimidating concepts into simple language and analogies.
3. **Quality over Speed**: Focuses on content quality and has a high standard.

Responsibilities:

1. Check that each material is accurate and has a verifiable source.
2. Reviews materials so that they are consistent and easy to understand
3. Use technical knowledge to give advice, inputs, and assist other members when needed.

Project Scope and Learning Goals

Target Learners

Our learning materials are designed for complete beginners with little to no experience in computer science. While they may know how to use AI for daily tasks, they don’t understand the logic of how it works.

Learning Goals

After using our materials, learners should be able to:

- Understand how algorithms, data structures, and connectionist AI work and are related to each other.
- Describe and explain examples of real world implementations of algorithms, data structures, and connectionist AI.

Content Outline

1. Data Structures & Algorithms

1.1. Foundations
1.2. Correctness and Complexity
1.3. Arrays & Dynamic Arrays
1.4. Linked Lists
1.5. Stack & Queues
1.6. Recursion & Divide-and-Conquer
1.7. Hash Tables
1.8. Trees, Binary Search Trees, and Tries
1.9. Heaps & Priority Queues
1.10. Algorithm Design Paradigms
1.11. Graphs
1.12. Specialized Structures

2. Connectionist AI

2.1. Foundations
2.2. The Artificial Neuron
2.3. Neural Network Architecture
2.4. How Neural Networks Learn
2.5. Evaluating Learning
2.6. Major Neural Network Types
2.7. Practical Implementation
2.8. Applications, Limitations, and Responsible AI

*Core concepts such as, basic data structures, common algorithms, and basics of connectionist AI will get more focus. So learners will have a better foundation on the material.

Content Plans

Shared Markdown Content

1. Serves as the source material that is used in the making of the study guide, PPT slides, and video.
2. Contains all explanations, examples, activities, references, and other assets of the learning materials.
3. Shared markdown content is going to be organized based on the content outline. Each topic will have its own file to represent the content. This ensures all material is well-organized and easy to use.

Study Guide and Test Bank

1. Serves as the deliverable with the most comprehensive explanations and content on the module.
2. Includes proper explanations, examples, and mini quizzes that help learners understand the material.
3. Test bank serves as a media for learners to test their knowledge and understanding after using the study guide.

Video Tutorials

Provides engaging and visual explanations on topic introductions and other topics, for example how recursions work. This will be shown through useful diagrams, graphs, and other visualisations.

PPT Slides

1. Serves as a more concise and visual focused learning material compared to the study guide.
2. Explanations for materials will be to be less text-focused, instead focusing on concise explanations and visual aids.

Live Mini Lecture

Live demonstration and mini lecture on breadth-first and depth-first search. Examples and demonstrations include maze logic.

Workflow and Coordination

1. Research

All research and investigation are going to be from books, research papers, educational websites, video explanations, AI, and other relevant sources. AI may be used to supplement or help in understanding certain materials.

2. Review

Markdown content is written after research has been done and is going to include, main explanations, relevant examples, references, and illustrations as needed. All markdown content must be checked and reviewed by one or more team members before being finalized.

3. Work Organization

- Github will be the main platform to store all related learning material files. This includes markdown content, assets, task lists, and other files.
- Contributions will be tracked through meaningful commits and recorded in team logs.
- Each task is ideally finished 2-3 days before official deadlines to make time for any corrections and revisions.

4. Communication

- WeChat is the main communication platform used for discussions, planning meetings, and team reminders.
- Team meetings are going to be conducted every week (most likely Fridays) to check team progress and discuss plans. Additional or emergency meetings will be done offline or online via WeChat calls as needed.
- Requests for reviews are notified through the WeChat group or directly done through Github. Reviews are done as soon as possible after the work is done.

6. Corrections

Corrections and revisions are done after content is reviewed. Requests for corrections are communicated with team members. Any changes in markdown content also means rechecking the relevant content in each deliverable. This ensures content is accurate and consistent with the source material.

7. Testing

- In between Stage 1 and Stage 2 (October 24 - November 7), for mini lecture rehearsals.
- Activity testing and feedback collection are going to be done periodically as production goes on.

8. Handling Obstacles

- **Unfamiliar Tools**
  - Consult team members or individuals who are familiar with the tool.
  - Familiarise ourselves by searching for guides or testing the tools.
- **Difficult Content**
  - Discuss with team members or teacher.
  - Look for more basic forms of content, in forms of YouTube videos, simplified guides, and others.
  - Consult AI to help with understanding materials.
- **Uneven Workloads**
  - Raise the issue from the start of workload assignment.
  - Communicate with team members to find a resolution or middle-ground
- **Delayed Tasks**
  - Prevent procrastination by setting clear internal deadlines.

Task List

A. Project Task List

**All dates are not final and are subject to change.*

No.	Task	Person in Charge	Internal Deadline	Status
1	**Preparation**			
1.1	Divide workload and agree upon team roles	All Members	19/10/2026	Done
1.2	Create Github repository and organize files	Jason (CIO)		
1,3	Create project plan and task list	All Members	20/10/2026	
2	**Markdown Development**			
2.1	Research all topics	All Members	-	Done
2.2	Complete markdown content V1.0		24/10/2026	
2.3	Complete markdown content V2.0		31/10/2026	Not Started
2.4	Internal review	Kenneth (Quality Lead), Winrich (CEO), All Members	1/11/2026	
3	**Production**			
3.1	Create study guide & test bank draft	All Members	2/11/2026	Not Started
3.2	Create PPT slides draft	Owen (Visual Design Lead)		
3.3	Create video plan and draft	Raphael (Video Lead)		
3.4	Create mini lecture script and materials	All Members		
3.5	Complete study guide and test bank V1.0		5/11/2026	
3.6	Complete PPT slides V1.0			
3.7	Complete video V1.0			
3.8	Complete peer review		13/11/2026	
4	**Finalisation**			
4.1	Complete mini lecture plans	All Members	13/11/2026	Not Started
4.2	Rehearse live mini lecture		17/11/2026	
4.3	Complete final markdown content		23/11/2026	
4.4	Complete final study guide and test bank			
4.5	Complete final PPT slides			
4.6	Complete final video			
4.7	Final review and revisions	Kenneth (Quality Lead), Winrich (CEO), All Members	25/11/2026	

B. Research Topic Distribution

No.	Topic Name	Person in Charge
1.1	Foundations	Jason (CIO)
1.2	Correctness and Complexity	
1.3	Arrays & Dynamic Arrays	
1.4	Linked Lists	
1.5	Stacks & Queues	Raphael (Video Lead)
1.6	Recursion & Divide-and-Conquer	
1.7	Hash Tables	
1.8	Trees, Binary Search Trees, and Tries	
1.9	Heaps & Priority Queues	Kenneth (Quality Lead)
1.10	Algorithm Design Paradigms	
1.11	Graphs	
1.12	Specialized Structures	Winrich (CEO)
2.1	Foundations	
2.2	The Artificial Neuron	
2.3	Neural Network Architecture	Michael (Content Lead)
2.4	How Neural Networks Learn	
2.5	Evaluating Learning	
2.6	Major Neural Network Types	Owen (Visual Design Lead)
2.7	Practical Implementation	
2.8	Applications, Limitations, and Responsible AI	

Thank You